🔑 **Answer Key — Teacher Copy · Lesson 5-5**

MSTAR PRACTICE WORKSHEET

Lesson 5-5: Determine the Volume of Rectangular Prisms

Grade 6 Mathematics · Standard 6.GR.2

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 SELECTED RESPONSE — CORRECT: C

What is the volume of a rectangular prism with l = 7 in, w = 3 in, h = 4 in?

A. 14 in³

B. 42 in³

C. 84 in³

D. 84 in²

V = l × w × h = 7 × 3 × 4 = 84 cubic inches.

Question 2 SELECTED RESPONSE — CORRECT: A

A time capsule box has a volume of 120 cm³. Its length is 10 cm and width is 4 cm. What is its height?

A. 3 cm

B. 4 cm

C. 6 cm

D. 12 cm

V = l × w × h → 120 = 10 × 4 × h → 120 = 40h → h = 3 cm.

Question 3 SELECTED RESPONSE — CORRECT: B

Which unit is used for volume?

A Square inches (in^2)

B Cubic inches (in^3)

C Inches (in)

D Degrees (°)

Volume measures 3D space, so it uses cubic units like in^3. Square units (in^2) are for area, and inches are for length.

Question 4 SELECTED RESPONSE — CORRECT: A

Two storage bins: Bin A is 8 × 6 × 5 inches. Bin B is 9 × 4 × 7 inches. Which holds more and by how much?

A Bin B by 12 in^3

B Bin A by 12 in^3

C Bin B by 24 in^3

D They hold the same

Bin A: 8 × 6 × 5 = 240 in^3. Bin B: 9 × 4 × 7 = 252 in^3. Bin B holds 12 in^3 more.

Question 5 SELECTED RESPONSE — CORRECT: D

A rectangular prism has length 8 cm, width 5 cm, and height 4 cm. What is its volume?

A 17 cm^3

B 80 cm^3

C 120 cm^3

D 160 cm^3

$V = l \times w \times h = 8 \times 5 \times 4 = 160\ cm^3$.

Question 6 SELECTED RESPONSE — CORRECT: B

A box has a volume of 72 in³. Its base is 6 in by 4 in. What is its height?

- A 2 in
- **B 3 in**
- C 12 in
- D 18 in

Height = V ÷ (l × w) = 72 ÷ (6 × 4) = 72 ÷ 24 = 3 in.

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7 TWO-PART QUESTION

Part A — correct answer: C

A rectangular time capsule container measures 8 inches long, 5 inches wide, and 3 inches tall. What is its volume?

- A 16 cubic inches
- B 40 cubic inches
- **C 120 cubic inches**
- D 158 cubic inches

V = l × w × h = 8 × 5 × 3 = 120 cubic inches.

- **A:** That adds 8, 5 and 3. Volume multiplies the three measurements.
- **B:** That is 8 times 5, the base alone. Multiply by the height of 3 as well.
- **D:** That is the SURFACE AREA of the six faces, in square inches. Volume multiplies 8 by 5 by 3.

Part B — correct answer: B

Which expression shows the same volume written as base area times height?

- A 2 × (8 × 5)
- **B (8 × 5) × 3**
- C (8 + 5) × 3
- D 8 × (5 + 3)

The base is the 8 by 5 rectangle, so B = 40 square inches, and V = B · h = 40 × 3 = 120 cubic inches.

Question 8 SELECT ALL THAT APPLY — CORRECT: A, B, C, E

Select ALL statements that correctly describe volume:

- **A** Volume is measured in cubic units.
- **B** V = l × w × h for a rectangular prism.
- **C** Volume can also be found as base area × height.
- **D** Volume is measured in square units.
- **E** Doubling only the height of a prism doubles its volume.

A, B, C, and E are true. D is false — square units measure area; volume fills space and uses cubic units.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9 WRITTEN RESPONSE · 2 POINTS

A box needs to hold exactly 60 cubic inches. Give three different sets of whole-number dimensions that work. Which set would make the box closest to a cube shape? Why might that matter?

Model answer: Any three whole numbers whose product is 60 work, such as 1 × 6 × 10, 2 × 5 × 6, and 3 × 4 × 5. Of those, 3 × 4 × 5 is closest to a cube because its edges are nearest to equal, and a cube-like box uses less material for its surface while holding the same 60 cubic inches.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.

Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.